

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 3.3: Waves — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.3: Waves
Driving Question	DRIVING QUESTION: How do waves carry energy through our environment — and how can understanding their behaviour protect Kenyan communities, improve technology, and help us make sense of the world around us?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Is a Wave? Introducing the Anchoring Phenomenon	Students observed a series of demonstrations and media: ripples spreading across a basin of water after a stone is dropped (or a short video clip of Lake Victoria shoreline waves), a slinky being shaken to produce transverse and longitudinal pulses, and a short audio clip of a loud thunderclap recorded near Nairobi. In each case students noted that a disturbance travels outward from its source without the medium itself travelling with it — a floating bottle-cap on the water surface bobs up and down but does not drift forward with the wave.	Students learned the defining characteristics of a wave: a wave is a disturbance that transfers energy from one point to another through a medium (or, for electromagnetic waves, through a vacuum) without net transfer of matter. They identified and labelled key wave anatomy — crest, trough, amplitude (A), wavelength (λ), period (T), and frequency (f) — on both transverse waves (rope, water surface) and longitudinal waves (sound, slinky compressions and rarefactions). They distinguished mechanical waves (require a medium; e.g., sound, water waves) from electromagnetic waves (travel through a vacuum; e.g., light, radio). Pedagogical note: ensure learners articulate the 'energy not matter' principle using the bottle-cap analogy before moving on; this is the conceptual anchor for the entire sub-strand.	The observed behaviour is explained by the nature of particle interactions in the medium. Each particle is coupled to its neighbour by intermolecular forces; when one particle is displaced it exerts a restoring force that passes the disturbance along. Because the particles return to their equilibrium positions, no net mass transport occurs — only energy propagates. For transverse waves (e.g., a shaken rope or Lake Victoria surface ripple), particle oscillation is perpendicular to the direction of wave travel. For longitudinal waves (e.g., the Nairobi thunderclap), particle oscillation is parallel to the direction of travel, creating alternating compressions (high pressure) and rarefactions (low pressure). The anchoring phenomenon — waves in Kenyan environments carrying energy — is thus explained at the particle level. Teachers should cold-call at least three students to re-explain the bottle-cap observation using the particle model before closing the lesson.
Lesson 2	Students observed a ripple tank (physical or	Through guided derivation, learners	The wave equation $v = f\lambda$ is not an empirical

The Wave Equation — Connecting Speed, Frequency and Wavelength	simulated) in which the frequency of the dipper was varied systematically while wave speed remained constant (same water depth). They recorded that increasing frequency produced shorter wavelengths, and decreasing frequency produced longer wavelengths. A class data table was built on the board, plotting f against λ and noting the product $f \times \lambda$ stayed approximately constant for each trial. A Kenyan radio context was introduced: an FM station in Nairobi broadcasts at 98.4 MHz — students were asked to predict the wavelength before deriving the equation.	established that in one period T a wave travels exactly one wavelength λ, giving $v = \lambda/T$. Substituting $T = 1/f$ yields the wave equation $v = f\lambda$. Students practised rearranging to find $f = v/\lambda$ and $\lambda = v/f$. They applied the equation to Kenyan contexts: calculating the wavelength of a 98.4 MHz Nairobi FM signal ($v = 3 \times 10^8$ m/s, $\lambda \approx 3.05$ m); calculating the frequency of a 340 m/s sound wave with $\lambda = 0.5$ m; and calculating the speed of a coastal wave near Mombasa given its frequency and measured wavelength. Pedagogical note: insist learners write the formula, substitute with units, and state the unit of the answer — unit tracking is a frequent exam error.	coincidence but a geometric necessity: in one full oscillation cycle (time = T), the wavefront advances by exactly one spatial cycle (distance = λ). Speed is distance ÷ time, so $v = \lambda/T = f\lambda$. This explains why, in a given medium (fixed v), a higher-frequency source must produce shorter wavelengths — the wavefronts are packed more closely together. The Nairobi FM example grounds abstraction in a familiar Kenyan technology. Teachers should use wait time (minimum 30 seconds) after posing each rearrangement problem before cold-calling, allowing all learners to attempt the algebra. Connect back to the driving question: knowing $v = f\lambda$ allows engineers to design antennae and predict where radio signals will reach across Kenya's varied terrain.
Lesson 3 Reflection of Waves: Why Does Sound Bounce Back — and Could It Explain the Fisherman's Puzzle?	Students observed: (1) a ripple-tank wavefront striking a straight barrier — incident and reflected wavefronts were traced and angles measured; (2) a clap produced near a large concrete wall on the school compound, listening for a distinct echo; (3) a short demonstration (or video) of a bat-echo analogy using a torch and mirror — the angle of the reflected beam equalling the angle of incidence. A Kenyan fisherman scenario was introduced as the anchoring phenomenon extension: fishers on Lake Victoria sometimes report hearing their own shouts reflected off distant cliffs — students were asked what property of waves explains this.	Sound reflects off hard surfaces; the angle of incidence equals the angle of reflection (the law of reflection). Echoes are reflected sound waves; reverberation occurs when multiple reflections overlap in an enclosed space. Students learned to draw normal lines, measure angles of incidence (θ_i) and reflection (θ_r), and confirm $\theta_i = \theta_r$ for plane wave reflection. They applied the echo formula: distance = $(v \times t) / 2$, and solved problems such as: a fisher hears an echo 0.6 s after shouting ($v_{\text{sound}} = 340$ m/s) — how far is the cliff? (answer: 102 m). Pedagogical note: many learners confuse the normal with the surface; spend explicit time drawing and labelling the normal at the point of incidence before measuring angles.	Wave reflection occurs because when a wavefront meets a boundary with a denser or rigid medium, the wave cannot continue and the energy is returned into the original medium. The law of reflection ($\theta_i = \theta_r$) follows from Huygens' principle: each point on the wavefront acts as a secondary source, and the geometry of secondary wavelet construction enforces equal angles. Hard, smooth surfaces (concrete walls, cliff faces, metal roofing sheets common in Kenyan urban buildings) are good reflectors; soft, porous materials (curtains, grass thatch) absorb energy and reduce reflection. The fisherman's puzzle is fully explained: sound energy from his shout travels outward, strikes the distant cliff (a rigid boundary), and reflects back to his ears as an echo. Teachers should connect this to real Kenyan applications pre-empted for Lesson 6: sonar used by Kenya Fisheries on Lake Victoria exploits this exact principle.
Lesson 4 Refraction of Waves — Why Does the Water Bend at the Shore and the Radio Fade in the	Students observed: (1) a ripple tank with a submerged perspex sheet creating a shallow-water region — wavefronts visibly slowed and bent (changed direction) as they crossed from deep to shallow water;	Learners learned that refraction is the change in direction of a wave caused by a change in its speed when it crosses a boundary between two media. The wave slows down in shallower water (or denser	Refraction is explained by the change in wave speed at a boundary. Wave frequency remains constant across boundaries (the source frequency does not change), so if v decreases, λ must also decrease ($v = f\lambda$).

Rift Valley?	(2) a photograph or video of ocean waves near the Kenyan coast (Diani Beach) bending as they approach the shore at an angle, with wavefronts turning to become more parallel to the shoreline; (3) a class discussion prompt: why do some communities in the Rift Valley report poor radio reception, especially for certain stations? Students predicted before explanation.	medium) and speeds up in deeper water (or less dense medium). The side of the wavefront that enters the new medium first slows (or speeds) first, causing the wavefront to pivot. Key rule: when a wave enters a slower medium at an angle, it bends toward the normal; into a faster medium, it bends away. Students applied this to: coastal wave refraction at Diani Beach (explains why waves always seem to hit the beach nearly head-on regardless of wind direction); atmospheric radio-wave refraction (explains signal fading in the Rift Valley where air density layers redirect radio waves). Pedagogical note: use the 'marching-soldiers' analogy — a row of soldiers walking from firm ground onto mud; those who hit the mud first slow down, causing the line to pivot.	The wavelength compression on one side of the wavefront causes the wavefront to rotate toward the slower medium. At Diani Beach, seabed rises gradually toward shore, progressively slowing coastal waves; each part of the wavefront slows at a rate proportional to local depth, causing the entire wavefront to swing parallel to the shore — this is why shore-break waves hit the beach almost head-on even when offshore swells arrive at an angle. For radio waves in the Rift Valley, temperature and humidity gradients create layers of differing air density and thus differing propagation speed; waves refract unpredictably, creating dead zones. Teachers should ask students (cold-call three) to explain the Rift Valley radio phenomenon in their own words using the speed-change mechanism before closing.
Lesson 5 Diffraction of Waves — Sound Around the Corner and Signals Over the Hill	Students observed ripple-tank wavefronts diffracting around obstacles and through apertures, confirming that diffraction is most pronounced when the gap width or obstacle size is comparable to the wavelength of the wave.	Students constructed wavefront diagrams showing that waves spread (diffract) around obstacles and through gaps, and that diffraction is greatest when the wavelength is approximately equal to (or larger than) the gap or obstacle size. They applied this principle to: (a) sound diffraction — sound waves have wavelengths of centimetres to metres, so they easily diffract around doorways, furniture, and corners in Kenyan classrooms and markets; (b) radio wave diffraction — long-wavelength KBC AM/FM signals diffract over hills, providing coverage in remote Kenyan valleys, whereas short-wavelength TV signals diffract poorly and are blocked by terrain, requiring relay stations; (c) ocean wave diffraction — waves diffracting around headlands (e.g., Ras Serani near Mombasa) can focus energy on sheltered beaches. Pedagogical note: require learners to draw at least two labelled wavefront diagrams (one showing little diffraction through a wide gap, one showing strong diffraction through a narrow gap) before moving to applications.	Diffraction is explained by Huygens' principle: every point on a wavefront acts as a new point source of secondary wavelets. When a wavefront passes through a narrow gap, only a small portion of the wavefront remains, and the secondary wavelets at the edges radiate outward into the geometric shadow, producing spreading. When the gap is much wider than λ, the secondary wavelets from the edges are a tiny fraction of the total wavefront and their effect is negligible — the wave continues in a straight beam with only minor edge spreading. When $\text{gap} \approx \lambda$, edge effects dominate and the wave fans out in all forward directions. This explains the Aberdare scenario: KBC longwave signals ($\lambda \sim$ hundreds of metres) easily diffract over the hill; TV signals ($\lambda \sim$ centimetres) travel in near-straight lines and are blocked. Teachers should connect this forward to Lesson 7: diffraction of ocean waves around coastal structures can unexpectedly concentrate wave energy on supposedly sheltered beaches — a critical environmental mitigation insight.

Lesson 6 Applications of Wave Properties in Real Life	Students observed a carousel of application demonstrations and/or short video clips: (1) a sonar/ultrasound animation showing a pulse sent from a vessel and the echo timed to calculate depth — linked to Kenya Fisheries vessels on Lake Victoria; (2) a medical ultrasound image (e.g., of a foetus) from Kenyatta National Hospital, showing how reflected sound pulses build an image; (3) a microwave oven cut-away diagram showing how microwaves reflect and refract inside the cavity to heat ugali or a chapati evenly; (4) a diagram of optical fibre communication used in Kenya's National Broadband network (the undersea cable at Mombasa) — total internal reflection carrying data as light pulses; (5) a seismograph trace from the Kenya Meteorological Department showing P-waves and S-waves from a tremor near the Rift Valley.	Each real-life Kenyan application deliberately exploits one or more wave properties. SONAR and ultrasound use REFLECTION — a directed pulse is sent out, the reflected echo is timed, and $\text{distance} = (v \times t)/2$ is applied. Medical ultrasound at KNH uses the same principle to image internal tissues. Microwave ovens exploit the specific frequency at which water molecules resonate (absorption) and reflection from metal walls to distribute energy. Fibre-optic cables (used in Kenya's TEAMS undersea cable terminating at Mombasa) exploit total internal reflection — light repeatedly reflects inside the glass fibre, carrying data over thousands of kilometres with minimal loss. Seismology uses the refraction and reflection of P- and S-waves through Earth's interior to map geological structures — critical for Kenya's Rift Valley geothermal energy exploration (KenGen). Pedagogical note: for each application, ask learners to identify WHICH wave property (reflection, refraction, diffraction, or a combination) is being exploited before revealing the explanation.	The unifying explanation across all applications is that engineers and scientists do not fight wave behaviour — they engineer systems that harness it. Reflection is exploited wherever distance measurement or imaging is needed (sonar, ultrasound, radar). Refraction is exploited in lens design and in understanding how radio signals propagate through the atmosphere over the Kenyan highlands. Diffraction is exploited in antenna design — a dish antenna is sized so that the target wavelength diffracts minimally, keeping the beam tight (microwave links) or, conversely, antenna arrays are designed for maximum diffraction for broad-area FM coverage across rural Kenya. Total internal reflection (a consequence of refraction at critical angle) enables fibre-optic internet infrastructure. Students should be able to state: 'In [application], the [property] of waves is exploited because [mechanism].' Teachers should cold-call at least five students using this sentence frame before closing the lesson.
Lesson 7 Effects of Waves on the Environment & Mitigation	Students observed: (1) photographs and/or a short documentary clip of coastal erosion at Malindi and Kilifi — showing how wave action undercuts cliffs and removes beach sand; (2) a diagram or video of the 2004 Indian Ocean tsunami's effect on the East African coast, discussing how long-wavelength, low-amplitude deep-ocean waves became high-amplitude destructive waves near shore; (3) a sound-level meter reading taken inside and outside a corrugated-iron-roofed classroom during a rainstorm — quantifying noise pollution from wave energy (sound); (4) photographs of Nairobi construction sites using vibration-dampening foundations to protect neighbouring buildings from seismic wave energy.	Each environmental impact is caused by a specific, nameable wave property: diffraction concentrates wave energy on unprotected shorelines (waves diffracting around a harbour breakwater can converge on the sheltered side); high-amplitude waves (tsunamis, storm surges) carry enormous mechanical energy that erodes coasts (Malindi, Watamu); seismic waves (P- and S-waves from Rift Valley tremors) transfer energy through Earth's crust, damaging poorly constructed buildings. Mitigation strategies are matched to the causative property: breakwaters and mangrove restoration reduce wave amplitude before waves reach the shore; early-warning buoys (Indian Ocean Tsunami Warning System, to which Kenya subscribes) detect anomalous wave patterns and trigger alerts; building codes in Nairobi require seismic-resistant designs in	The environmental effects of waves are explained by the same physics principles studied in Lessons 1–5. Coastal erosion at Malindi results from the kinetic energy carried by waves (proportional to amplitude squared) being transferred to the coastline. Tsunami destructiveness results from shoaling — as the wave enters shallow coastal water, its speed decreases (refraction/shoaling), wavelength compresses, but energy is conserved, so amplitude increases dramatically. Diffraction of ocean waves around engineered structures (harbour walls) can inadvertently redirect wave energy into previously sheltered areas — explaining why some mitigation structures create new erosion problems if poorly designed. Seismic wave damage is greatest where S-wave frequency matches the natural resonant frequency of buildings (a wave

		earthquake-prone zones near the Rift; sound-absorbing materials reduce noise pollution in urban schools. Pedagogical note: students often conflate 'the wave is dangerous' with 'all waves are dangerous' — stress that the same reflection/diffraction/refraction principles that cause harm are also used in mitigation technology.	superposition/resonance effect previewed here). Effective mitigation requires understanding the wave property causing the problem: you cannot design a breakwater without understanding diffraction; you cannot site a sea wall without understanding refraction. Teachers should close with the driving question: 'How does understanding wave behaviour protect Kenyan communities?' — requiring three students to give specific, property-linked answers.
Lesson 8 Model Building, Numerical Consolidation & DQB Closure — How Wave Science Keeps Kenyan Communities Safer	Students reviewed their Driving Question Board entries accumulated across all seven lessons, physically (or digitally) moving sticky notes to mark questions that had been answered, identifying any remaining gaps. They then observed their own cumulative wave models — diagrams built and revised lesson by lesson — and compared their Lesson 1 initial model (often showing waves as 'moving water') with their current model (wavefronts, particle oscillation, energy transfer, labelled properties). A final consolidation problem set was distributed, containing five multi-step numerical problems drawn from Kenyan contexts: (1) Nairobi FM wavelength; (2) Mombasa sonar depth calculation; (3) echo-distance for a KBC engineer testing a radio tower near Nakuru; (4) coastal wave refraction angle change at Diani; (5) diffraction gap-width decision for a rural antenna installation in Turkana County.	Students learned: the wave equation $v = f\lambda$ and how to use it to calculate speed, frequency, and wavelength in Kenyan contexts (Nairobi FM radio, Mombasa sonar, KBC transmission towers); how reflection is used in sonar and ultrasound (distance = $vt/2$); how refraction explains coastal wave bending and Rift Valley radio dead zones; how diffraction explains why long-wave KBC signals reach remote valleys but TV signals do not; how these properties combine to cause and mitigate environmental hazards (Malindi erosion, Indian Ocean tsunamis, Rift Valley seismic risk). They also reviewed SI units for all wave quantities (m, Hz, m/s, s) and practised unit analysis as an exam strategy. Pedagogical note: use the five-problem consolidation set as a formative assessment — circulate and note which problems reveal persistent misconceptions (unit errors, formula inversion, confusion of amplitude with wavelength) and address these explicitly before the summative assessment.	The final explanatory synthesis connects every lesson back to the driving question. Waves carry energy — not matter — through our environment via particle interactions (Lesson 1). The speed, frequency, and wavelength of a wave are related by $v = f\lambda$ (Lesson 2). When waves meet boundaries, they reflect (Lesson 3), enabling sonar, ultrasound, and echo-ranging. When wave speed changes at a boundary, waves refract (Lesson 4), bending coastal waves at Diani and causing radio dead zones in the Rift Valley. When waves pass obstacles or gaps comparable to their wavelength, they diffract (Lesson 5), allowing KBC signals to reach remote Kenyan valleys but also focusing wave energy on unprotected shorelines. These properties are harnessed in technology (Lesson 6) and must be understood to design effective environmental protections (Lesson 7). The cumulative student model should now show: a labelled wavefront diagram including arrows for energy direction, particle oscillation arrows (transverse vs longitudinal), boundary interactions (reflection, refraction, diffraction), and at least two annotated Kenyan real-world application examples. Teachers should display two or three exemplary student models and facilitate a class discussion on what makes a model scientifically complete before formally closing the DQB.